

3 (**short**) A transformer with a primary:secondary turns ratio of 10:1 has a load of impedance $(0.3 + j0.2) \Omega$ connected to its secondary winding and its primary winding is connected to a 240 V supply. The equivalent circuit parameters of the transformer referred to the primary are:

$$R_1 = 6 \Omega, R_2' = 4 \Omega, X_1 = 7 \Omega, X_2' = 3 \Omega.$$

The iron loss resistance R_0 and magnetising reactance X_0 are large enough to be neglected.

- (a) Calculate the impedance of the load referred to the primary. [2]
- (b) Calculate the load current. [3]
- (c) Calculate the real and reactive power in the load. [3]
- (d) Calculate the transformer efficiency. [2]